ABOUT US

Gravity was founded in South Africa in 1999. Since then Gravity has been a leader in the work at height industry. Assisting in developing a variety of standards, providing excellent training, producing innovative solutions for work sites as well as a range of fall arrest equipment.

Gravity is committed to providing quality services and products with our clients satisfaction being our highest priority.

INDEX

1. What is radiation?

There are four major types of radiation: alpha, beta, neutrons, and electromagnetic waves

Any form of energy that moves through space is a form of radiation. Below are a few examples of some typical sources of different types of radiation.

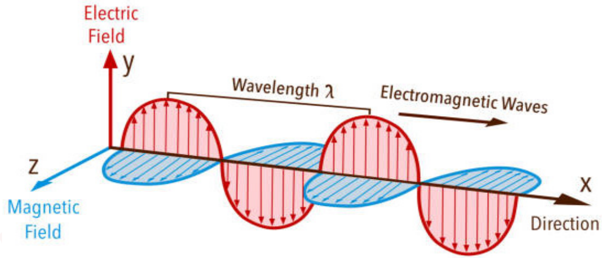
Light radiation

Acoustic radiation

Heat radiation

Although there are many forms of radiation such as the types mentioned above and many more.

2. What is electromagnetic radiation?



This course will be focussing specifically on electromagnetic radiation or EMR. Electromagnetic radiation propagates in the form of an electromagnetic wave (EMW).

An electromagnetic wave is created by a change in an electric field which creates a magnetic field and vice versa.

The frequency is determined by how many full cycles a wave can complete in one second.
The amplitude is the value given to describe how strong (how much energy it carries) a wave is.

It is important to understand that the intensity of the RF Waves (EMW) decrease as it propagates away from the source.

X-rays, light bulbs, the sun, fire, microwave ovens etc. all emit EMW's.

3. Introduction to ICNIRP guidelines.

"ICNIRP known as **International Commission on Non-Ionizing Radiation Protection**. They provides scientific advice and guidance on the health and environmental effects of non-ionizing radiation (NIR) to protect people and the environment from detrimental NIR exposure.

Non-ionizing radiation refers to electromagnetic radiation such as ultraviolet, light, infrared, and radiowaves, and mechanical waves such as infra- and ultrasound. In daily life, common sources of NIR include the sun, household electrical appliances, mobile phones, Wi-Fi, and microwave ovens.

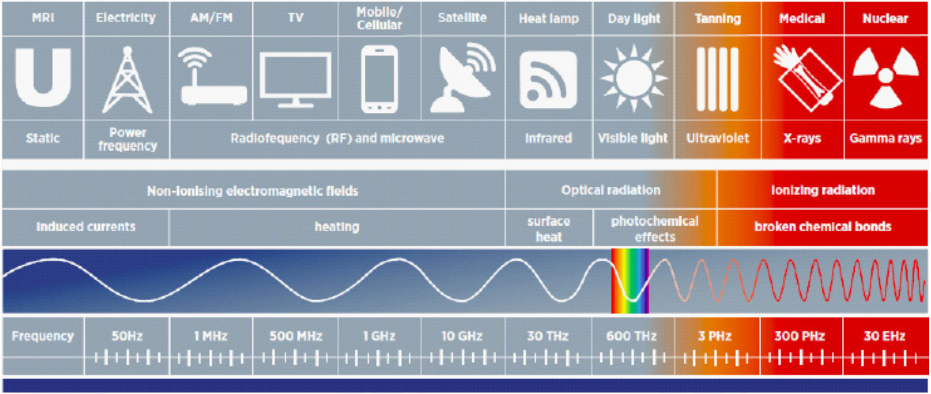
Within the electromagnetic spectrum, NIR is situated below the ionizing radiation band that includes x-rays. NIR has less energy than ionizing radiation and cannot remove electrons from atoms, i.e. NIR can not ionize (except for part of the UV band). NIR is sub-grouped into different frequency or wavelength bands. The different subgroups have different effects on the body and require different protection measures.

ICNIRP gives recommendations on limiting exposure for the frequencies in the different NIR subgroups. It develops and publishes Guidelines, Statements, and reviews used by regional, national, and international radiation protection bodies, such as the World Health Organization. ICNIRP is a main contributor to the international scientific NIR dialogue and the advancement of NIR protection."

For RF waves the ICNIRP guidelines set two limits: 1) Public Exposure Limit (Yellow exclusion zone) and 2) Occupational Exposure Limit (Red Exclusion zone)

https://www.icnirp.org/

The electromagnetic spectrum



Radio frequencies (RF) range from 100 kHZ to 300 GHz, and as such are invisible to the human eye, but this does not mean that they do not affect us.

Most of our current technologies operates in the RF part of the spectrum.

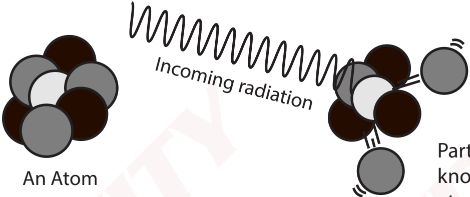
Sources of RF radiation include:

- Laptops
- Electric razors
- LED/LCD monitors

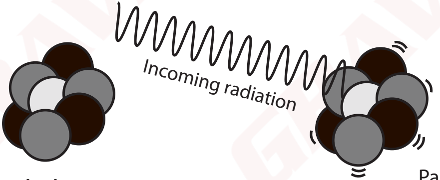
4. Ionizing vs. non-ionizing radiation

Radiation is divided into two categories, ionizing and non-ionizing.

If the incoming radiation from a source has enough energy the radiation can CHANGE THE COMPOSITION OF AN ATOM by knocking out an electron. Or in other words the radiation IONIZES the atom. This can lead to increased chances of diseases such as cancer. Thus this is referred to as IONIZING radiation.



Now, if the incoming radiation does NOT have enough energy to remove particles from an atom, then the atom only HEATS UP, thus NOT IONIZING the atom. This can cause burns, heatstroke or organ failure, **but only at a very high intensity**. Thus this is referred to as NON-IONIZING radiation. For low intensity RF exposure (ie exposure below the ICNIRP public limits) there are no health risk.



Radio frequency radiation is a form of non-ionizing radiation and it is important to understand the effects of exposure to this type of radiation can have on the human body and how to determine whether a person has been over-exposed to radiation.

5. Effects of non-ionizing radiation

There are two main health effects that can result when exposed to high levels of RF energy:

- **Heating of the human body**

Exposure to high levels of RF energy will result in a person experiencing symptoms similar to sunburn or heatstroke.

- **EMF interference**

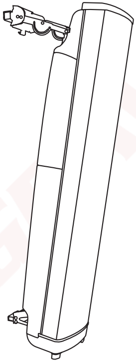
Any electromagnetic radiation can convert any conductive material such as metal implants or pacemakers into antennae. This can lead to either heating or malfunctioning of devices.

6. How to avoid exposure to RF radiation

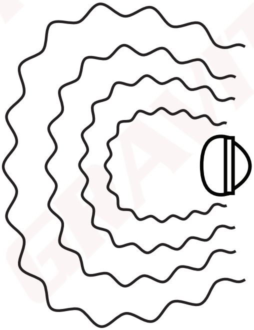
In order for workers to avoid RF over exposure:

1. Stepping away from the RF source.
2. Switching off the RF source before commencing of work
3. Adhering to the ICNIRP safety Standard.

Directional antennas

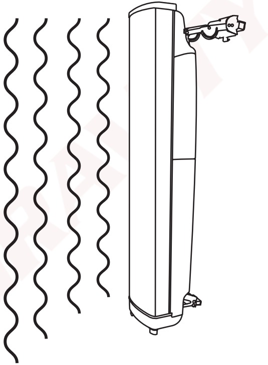


Top view



RF radiation does propagates out over a larger area, although it remains IN FRONT of the antenna.

Side view

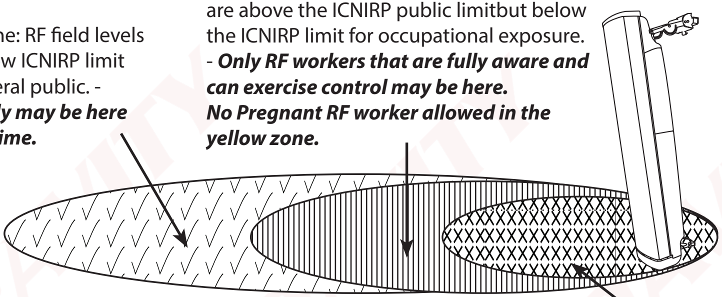


RF waves propagate outward away from the antenna and the intensity of the waves becomes smaller and smaller.

RF safety zones

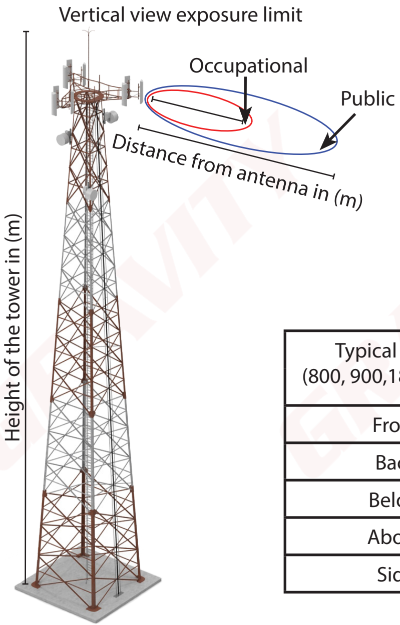
Blue zone: RF field levels are below ICNIRP limit for general public. - ***Anybody may be here at any time.***

Yellow/Public exclusion zone: RF field levels are above the ICNIRP public limitbut below the ICNIRP limit for occupational exposure.
- ***Only RF workers that are fully aware and can exercise control may be here.***
No Pregnant RF worker allowed in the yellow zone.



Red / Occupational exclusion zone: RF field levels are above occupational ICNIRP limit
- ***RF Workers are allowed to pass quickly through Red / Occupational Exclusion Zone, but not allowed to work in Red zone while antenna is transmitting RF waves.***

EMF distances from antenna

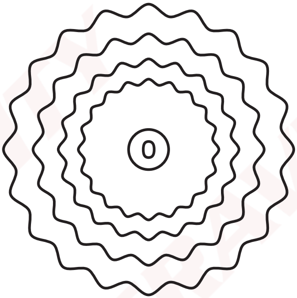


Typical Rural (800, 900)	Public (m/cm)	Occupational (m/cm)
Front	9,91 m	4.43 m
Back	28 cm	13 cm
Below	57 cm	26 cm
Above	46 cm	21 cm
Side	3.78 m	1.69 m

Typical Urban (800, 900,1800, 2100)	Public (m/cm)	Occupational (m/cm)
Front	11,49 m	5.14 m
Back	30 cm	13 cm
Below	76 cm	34 cm
Above	73 cm	32 cm
Side	4.59 m	2.05 m

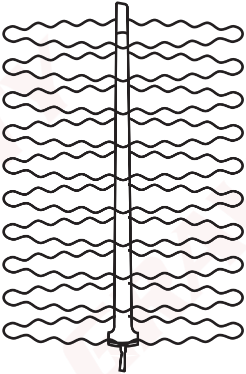
Omnidirectional antennas

Top view



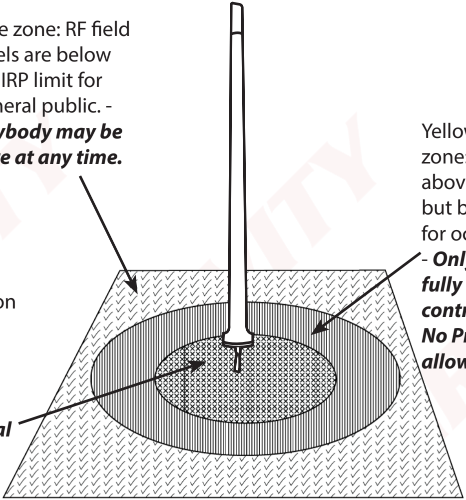
As shown here a worker will be exposed to RF radiation no matter where they stand.

Side view



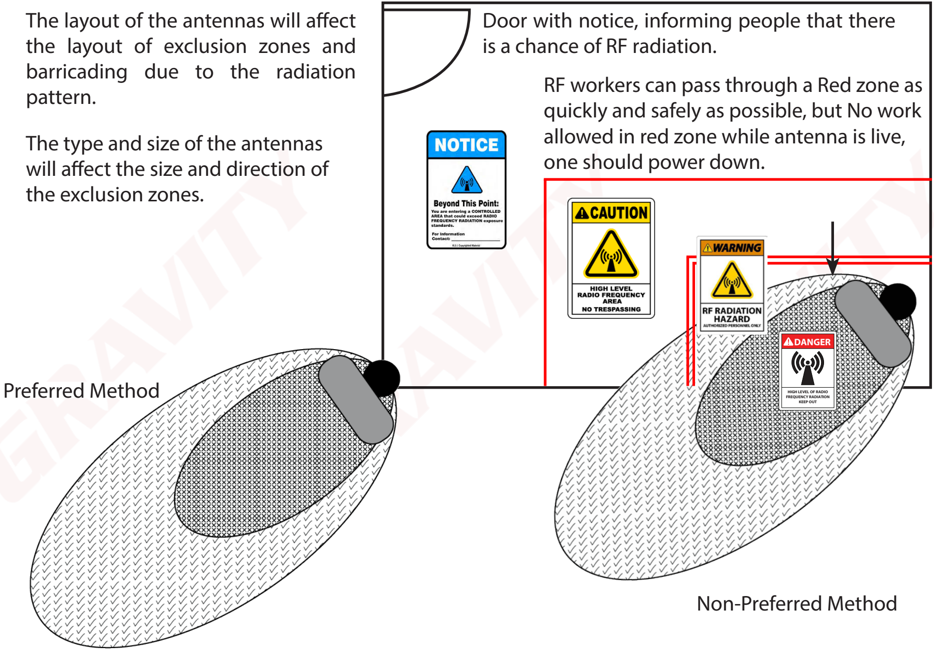
RF radiation propagates outward in all directions and the intensity becomes smaller and smaller.

RF safety zones



The layout of the antennas will affect the layout of exclusion zones and barricading due to the radiation pattern.

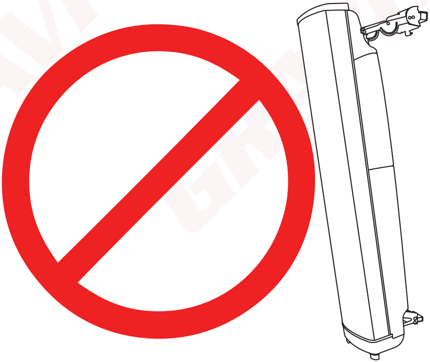
The type and size of the antennas will affect the size and direction of the exclusion zones.



This is only an estimate of the safety zones of an average directional antenna. The frequency and power of the antenna will cause these zones to shift and change. Therefore, **it is of utmost importance to know with which type of antenna you are working on**. See manufacturer's instructions and recommendations.

Although these zones must be marked and easily visible, a good rule of thumb is to NEVER WORK DIRECTLY IN FRONT of any RF antenna without following the relevant power-down procedures which must be obtained from the service provider. Confirm with the use of the RF Monitor that the area you are planning to work in is safe before you start the work.

Should you access a high RF exposure zone, RF workers can pass through a Red zone as quickly and safely as possible, but No work allowed in red zone while antenna is live, one should power down.



7. Basic Functions of an RF Scanners

As the RF scanner is moved through the area, RF monitors measures the RF exposure at specific location and for that specific time. It displays the results as a % of the ICNIRP Occupational limits.

1. The scanner can be switched between measure and monitor mode:
   - "Measure" is used to determine the strength of the field.
   - "Monitor" is used to constantly ensure that the levels of exposure do not change.
2. The scanner has 7 LED lights that light up one after the other in accordance with the measured intensity of the RF exposure.
3. If 4 of the 7 lights are lit, the radiation is approaching 100% of the exposure limit. 5 lights are exceeding 100%.
4. A good control measure is for us to avoid prolonged exposure in an area as soon as the scanner starts beeping.

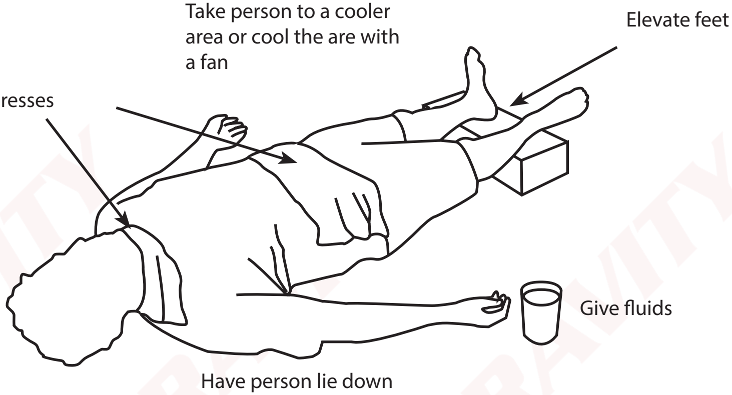
General RF safety monitor characteristics:* Operating Frequency bands* Operating modes (Probe architecture)
* Which RF safety limits referenced* Calibration validity

RF Scanner Indicators

LED's	Exposure Limit	Audio Alarm
7	250%	4 Hz
6	160%	2 Hz
5	100%	1 Hz
4	63%	1 Hz
3	40%	-
2	15%	-
1	5%	-

Any audio indicates that the level of RF radiation is above the recommended exposure limit as set by the ICNIRP, one should move away from the antennas or power down the antennas to continue working in that area.

8. Treatment of RF Over Exposure



- First Aider remove worker from exposure area to a cool environment and provide cool drinking water.
- Apply cold water or ice to burned areas.
- Seek immediate medical attention.
- Severe MW or RF overexposure may damage internal tissues without apparent skin injury, so a follow-up physical examination is advisable.

9. EMF Signage

Blue Zone Notice :

You are entering a Radio Frequency controlled environment.

- Most sites can use this sign at the gate as a pre-caution even if the levels on the ground are below the public limit.

Yellow Caution Zone

- States that “beyond this point” fields may exceed the Public safety limit.
- RF Workers may pass the sign but not the public.

Red Warning Zone

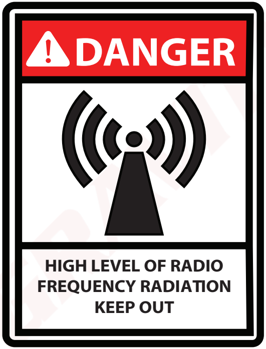
- Beyond this point RF fields exceed the Occupational safety limit. I.e RF workers is not allowed work in the red zone.
- Sites where some areas on the ground exceed the occupational limit should use this sign.

Examples of a Caution Signs:

- This sign represents IONIZING radiation and means that the area may not be entered by RF technicians without receiving the proper training.

Electromagnetic radiation can interfere with devices like pacemakers and should not be exposed to any radiation.

Examples of a Danger Signs:



- States that "beyond this point" fields exceed the Occupational safety limit.
- It means nobody should enter the area unless power is reduced.
- Example of high power broadcasting sites(Sentech Sites)
- Use a suitable RF monitor to confirm exposure levels before working on high power broadcaster sites.

All these signs are of utmost importance in order to maintain **ACCESS CONTROL**. The risk assessment can be as thorough as possible, but it would be useless if there was no control over who enters the site and when.

All telecommunication service providers have their own form of access control procedures and must be adhered to completely.

When on site the safe work procedures given by the service provider must be adhered to at all times.

10. In Conclusion:

In order for workers to remain safe within a RF radiation environment remember the following:

- Always follow the warning signs on the work site.
- Try to remain out of the radiation zones of any antennas.
- Do not expose yourself to RF radiation for long periods of time.
- Should you be unsure of radiation levels either evacuate the area or measure the levels and take the appropriate action.
- Always contact the RF safety officer for assistance